Altitude response of a free-falling permeable plate to a fluctuation of the gravitational acceleration

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Abstract

We quantify how a transient fluctuation of the gravitational acceleration displaces a two-dimensional free-falling permeable plate, using direct numerical simulations at a Galilei number $Ga = 150$. A plate free to rotate either gains or loses altitude, according to the order in which the vertical component g_y is raised and lowered. When the angular position is constrained, every case tested gains altitude, irrespective of that order. A fluctuation of the horizontal component g_x always produces a gain, between one and two orders of magnitude larger than for g_y. The results are limited to Darcy numbers $Da = 10^{-5}$ and 10^{-4} for the constrained plate, and the horizontal cases have not recovered their descent speed within the simulated window, so the reported gains are lower bounds.

1 Introduction

A body falling freely in a fluid does not respond to a transient perturbation of its weight by returning to its original trajectory. The perturbation leaves a residual displacement even when it carries no net impulse, and the sign and magnitude of that displacement are not obvious *a priori*. This report quantifies the residual altitude change of a permeable plate for fluctuations of the vertical and of the horizontal component of the gravitational acceleration.

All symbols are nondimensional unless carrying a hat, in which case they are dimensional. Lengths are scaled on the plate length c, velocities on the gravitational velocity $u_g \equiv (\hat{m}\hat{g}/\hat{\rho}\hat{c}^2)^{1/2}$, and the fluid density is unity.

2 Methodology

2.1 Governing parameters

The flow is governed by the Galilei number $Ga \equiv u_g c/\nu$, where ν is the kinematic viscosity; the Darcy number $Da \equiv K/c^2$, where K is the permeability; the nondimensional mass m; the chord-to-thickness ratio $\chi \equiv c/f$, where f is the thickness; the nondimensional moment of inertia I_O about the plate centre; and the offset e of the centre of gravity from that centre along the body normal. The solid volume is neglected, so buoyancy is null and the nondimensional gravitational acceleration is $g = 1/m$.

We take $Ga = 150$, $\chi = 10$, $m = 0.80$, $e = 1.16$ and $I_O = 1.15$ throughout, and vary Da between 10^{-5} and 10^{-3}.

2.2 Gravity fluctuation

The fluctuation is applied over a single period $T_G = 2$ convective times beginning at t_0, as a transient change of the gravitational acceleration,

$$\boldsymbol{g}_{\mathrm{eff}}(t) = \boldsymbol{g} \pm A\, g \sin\left[\frac{2\pi(t - t_0)}{T_G}\right] \boldsymbol{e}, \tag{1}$$

where A is the semi-amplitude as a fraction of g; $\boldsymbol{e}$ is the direction of the fluctuation, $\boldsymbol{e}_y$ for a fluctuation of the vertical component and $\boldsymbol{e}_x$ for the horizontal; and the sign selects the order in which the two excursions occur. The net impulse of equation (1) is null, so the fluctuation changes neither the mean nor the final gravitational force. For $A \geq 1.5$ applied vertically the effective gravity reverses during part of the period.

2.3 Numerical method

The incompressible Navier–Stokes equations are solved with the boundary data immersion method (BDIM) on a uniform Cartesian grid, in a frame translating with the plate. Permeability enters as a Brinkman resistance blended into the BDIM mobility, which recovers the impermeable body exactly as $Da \to 0$. Three degrees of freedom are integrated with the added-mass tensor treated implicitly. The plate length is 64 cells, giving 6.4 cells across the thickness, in a domain of $16c \times 16c$.

Each configuration is first advanced from rest to its terminal state, then branched: one continuation at constant gravity as a reference, and one per fluctuation. All branches of a configuration start from an identical saved state, so the difference between them is due to the fluctuation alone. The altitude change is

$$\Delta Y(t) \equiv Y(t) - Y_{\mathrm{ref}}(t), \tag{2}$$

where Y_{ref} is the altitude of the reference plate, so that $\Delta Y > 0$ denotes a gain.

3 Results

3.1 Plates free to rotate

A plate free to rotate either gains or loses altitude, and which of the two occurs is set by the order of the fluctuation (figure 1). Reducing g_y first produces a gain, and raising it first produces a loss, at every amplitude tested. The response is close to antisymmetric: the residual sum $\Delta Y_{\downarrow} + \Delta Y_{\uparrow}$ is -0.0025, $+0.0234$ and -0.0231 for $A = 0.5$, 1.5 and 2.5, against individual responses of order 0.1.

The outcome is therefore deterministic rather than uncertain. The sign of the residual follows the sign of the first excursion, and reversing the order reverses the displacement.

3.2 Plates with a fixed angular position

When the angular position is constrained, every plate tested gains altitude (figure 2). This holds at both Darcy numbers and at all three amplitudes, for both orders of the fluctuation: all twelve residuals are positive, between $+0.004$ and $+0.211$.

Constraining the rotation therefore removes the part of the response whose sign follows the order, and leaves a rectified part that displaces the plate upwards whichever way the fluctuation runs. Decomposing the residual into a component that changes sign with the order and one that does not, the former falls by a factor between three and eight when the rotation is constrained, while the latter rises from at most 0.012 to 0.167.

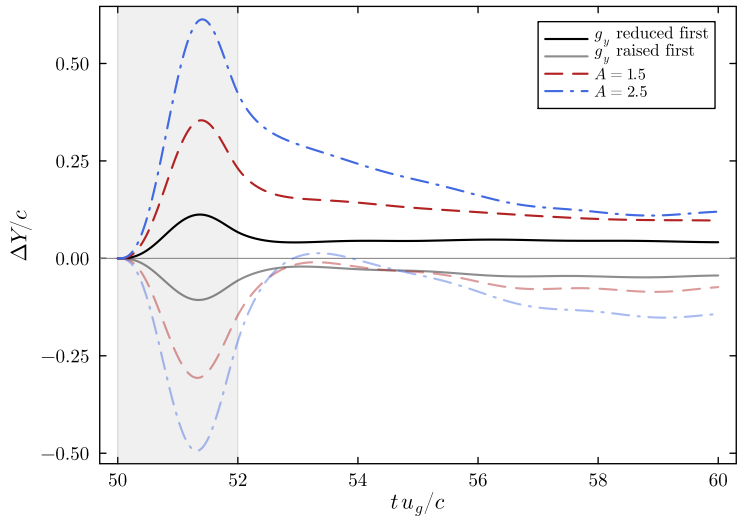


Figure 1: Altitude change versus time for a plate free to rotate, for values of the semi-amplitude A from 0.5 to 2.5 and both orders of the fluctuation, for $Da = 10^{-4}$. The shaded band is the fluctuation period.

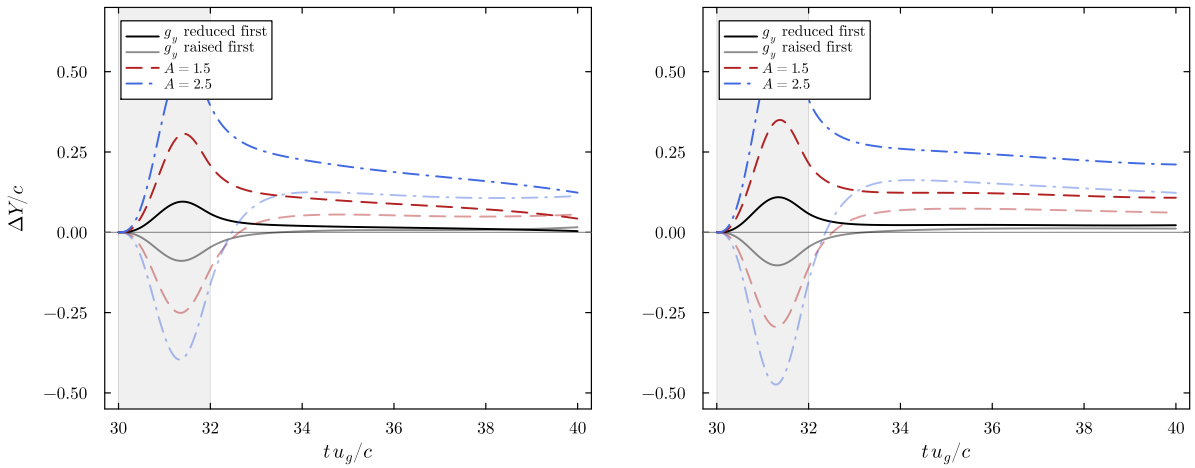


Figure 2: Altitude change versus time for a plate with a fixed angular position, for values of the semi-amplitude A from 0.5 to 2.5 and both orders of the fluctuation: (a) $Da = 10^{-5}$, (b) $Da = 10^{-4}$. The shaded band is the fluctuation period.

3.3 Fluctuation of the horizontal component

A fluctuation of g_x always produces an altitude gain, at every Darcy number and amplitude tested, and for both signs (figure 3). The vertical component of gravity is constant throughout these simulations, so the gain is produced entirely by the lateral forcing.

The gain is between one and two orders of magnitude larger than that produced by a fluctuation of g_y of the same amplitude (figure 4 and table 1). It is also rectified to a closer tolerance: at $Da = 10^{-4}$ and $A = 2.5$ the two signs give +2.119 and +2.106, a difference of 0.6%.

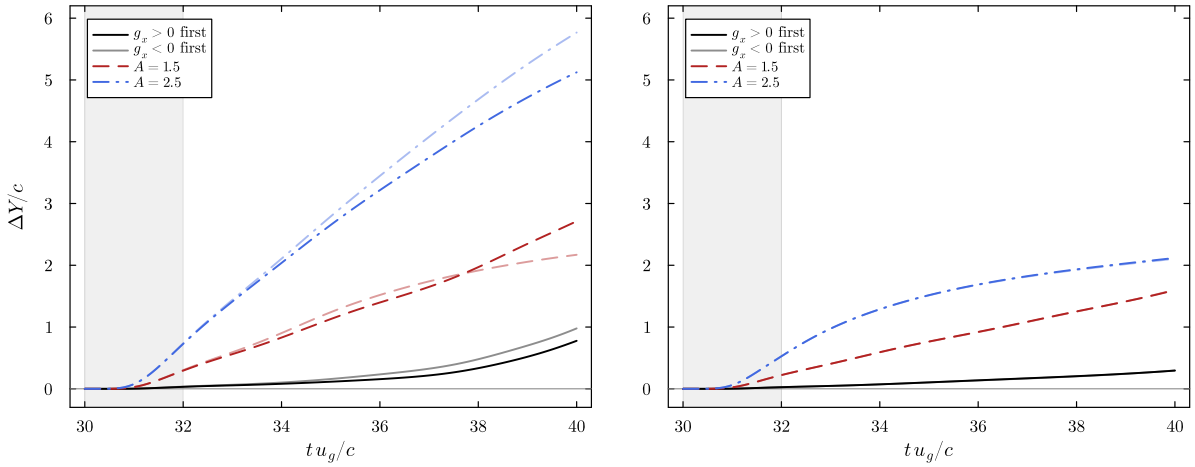


Figure 3: Altitude change versus time for a fluctuation of the horizontal component of the gravitational acceleration, for a plate with a fixed angular position, for values of the semi-amplitude A from 0.5 to 2.5 and both signs: (a) $Da = 10^{-5}$, (b) $Da = 10^{-4}$. The shaded band is the fluctuation period. Note the ordinate range, ten times that of figure 2.

Table 1: Altitude gain ten convective times after the onset of the fluctuation, averaged over the two orders, and the ratio between the horizontal and the vertical fluctuation, for a plate with a fixed angular position.

Da	A	$\Delta Y/c$, g_x	$\Delta Y/c$, g_y	ratio
10^{-5}	0.5	0.878	0.010	90
10^{-5}	1.5	2.441	0.049	50
10^{-5}	2.5	5.445	0.118	46
10^{-4}	0.5	0.297	0.017	18
10^{-4}	1.5	1.601	0.084	19
10^{-4}	2.5	2.112	0.167	13

4 Caveats

Four limitations bound the results.

Limited range of Da. The constrained-rotation results of sections 3.2 and 3.3 cover only $Da = 10^{-5}$ and 10^{-4}, a single decade. The statement that every constrained plate gains altitude is therefore established over that decade and not over the wider range 10^{-5} to 10^{-3} explored for the plate free to rotate.

Time histories not converged for a horizontal fluctuation. For a fluctuation of g_y the altitude change settles within the simulated window, and the descent speed returns to its

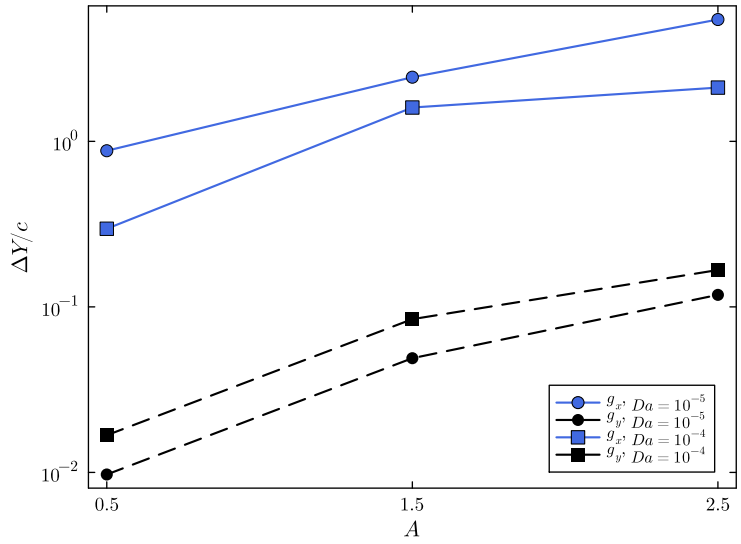


Figure 4: Altitude gain ten convective times after the onset of the fluctuation versus semi-amplitude, for fluctuations of the horizontal and of the vertical component of the gravitational acceleration, for $Da = 10^{-5}$ and 10^{-4}, for a plate with a fixed angular position.

reference value to within 0.5%. For a fluctuation of g_x it does not: ten convective times after the onset the plate is still descending between 7% ($Da = 10^{-4}$) and 38% ($Da = 10^{-5}$) more slowly than the reference, and ΔY is still growing. The gains of section 3.3 are therefore lower bounds, and the ratios of table 1 depend on the instant at which they are evaluated.

Mechanisms are not established. Only the trajectories and the integrated loads were recorded. No force decomposition, quasi-steady baseline or flow-field data was collected, so no mechanism is demonstrated and none is claimed here.

Numerical resolution. The Brinkman screening length $c\sqrt{Da}$ is 0.64 and 0.20 cells at $Da = 10^{-4}$ and 10^{-5}, so the internal layer is not resolved; for a thin plate the governing quantity is the pressure drop across it, which does not require it. No grid-convergence study was made. The comparisons are between simulations at identical resolution and are therefore robust to a common discretisation error, but absolute values are not established. In the most displaced case the plate moves 3.4 chords laterally in a domain 16 chords wide with periodic lateral boundaries, and interaction with its own periodic image is untested.

5 Conclusions

Three results follow from 47 simulations at $Ga = 150$.

A plate free to rotate either gains or loses altitude in response to a fluctuation of the vertical component of the gravitational acceleration, according to the order in which that component is raised and lowered.

A plate whose angular position is fixed gains altitude in every case tested, at both Darcy numbers and all three amplitudes, and for both orders.

A fluctuation of the horizontal component always produces a gain, between one and two orders of magnitude larger than for the vertical component at the same amplitude, evaluated ten convective times after the onset.

Code and data: https://github.com/ignaziomviola/FallingPlate2D.jl.